

Lab 1 - Linux Terminal To First Script

CST2415 Linux practical bootcamp | Brightspace submission lab

Purpose

This lab combines the course's early BASH shell, Linux directory structure, command-line file management, and first Bash script materials into one fast beginner lab. The main goal is not memorization. The goal is to use Linux to navigate, inspect, create, script, and explain.

What You Submit

- One Word, PDF, or clearly organized text file uploaded to Brightspace.
- Terminal output or screenshots showing your commands and results.
- The full contents of both scripts you create.
- Output from running both scripts.
- Written answers to the reflection questions in Part F.

Item	Evidence to include	Done
Part A	Identity/location commands	
Part B	Folder and file commands	
Part C	File viewing/search commands	
Part D	hello.sh contents and output	
Part E	system_snapshot.sh contents and output	
Part F	Written reflection answers	

Setup

Open a Linux terminal. Fedora is preferred, but another Linux shell is acceptable. If your environment is not working, follow the instructor demo and document where your setup failed.

```
whoami
hostname
date
pwd
ls
```

Part A - Prove Where You Are

Run the commands below. Copy the output or capture a screenshot.

```
whoami
hostname
date
pwd
ls
```

Answer in your submission:

- What user are you logged in as?
- What folder were you in when you ran pwd?
- What happened when you ran ls?

Part B - Create A Working Folder

Create a lab folder and move into it.

```
mkdir -p ~/cst2415_lab1
cd ~/cst2415_lab1
pwd
ls -la
```

Create a notes file.

```
echo "Today I practiced Linux commands and Bash scripting." > notes.txt
cat notes.txt
ls -l
```

Part C - Navigate And Inspect Files

Practice the same ideas from the Linux directory structure materials: current directory, parent directory, absolute path, and relative path.

```
pwd
cd ..
pwd
cd ~/cst2415_lab1
pwd
cd /
pwd
cd ~/cst2415_lab1
```

Create a second file and inspect both files.

```
echo "Linux is case sensitive." > commands.txt
echo "date and Date are different command names." >> commands.txt
cat commands.txt
file notes.txt
head notes.txt
tail notes.txt
grep Linux commands.txt
grep -i linux commands.txt
```

Part D - Create Your First Bash Script

Create a script named hello.sh. You may use nano, vi, or another terminal editor.

```
nano hello.sh
```

Put this inside the file:

```
#!/bin/bash
echo "Hello from my first Linux script"
echo "The current date is:"
date
echo "I am currently in:"
pwd
```

Save the file, make it executable, and run it.

```
chmod +x hello.sh
ls -l hello.sh
./hello.sh
```

Part E - Build A Useful System Snapshot Script

Create a second script that collects beginner system information. This connects Linux commands to software, support, analysis, and troubleshooting work.

```
nano system_snapshot.sh
```

Put this inside the file:

```
#!/bin/bash
echo "=== System Snapshot ==="
echo "User: $(whoami) "
echo "Host: $(hostname) "
echo "Date: $(date) "
echo
echo "=== Linux Version ==="
cat /etc/os-release | head -5
echo
echo "=== Disk Space ==="
df -h | head
echo
echo "=== Memory ==="
free -h
```

Run it and save the output.

```
chmod +x system_snapshot.sh
./system_snapshot.sh
./system_snapshot.sh > snapshot_output.txt
cat snapshot_output.txt
```

Part F - Writing Component

Answer in complete sentences. Short answers are fine, but they should show that you understand what happened.

- What is the difference between a terminal and a shell?
- Explain the difference between an absolute path and a relative path. Give one example of each from today's lab.
- Why did we run `chmod +x` before using `./hello.sh`?
- What did `system_snapshot.sh` collect? Name at least three pieces of information.
- How could a beginner developer, support technician, cybersecurity analyst, or data worker use a script like this?

If Your Environment Failed

Do not fake output. Submit what happened, including any error message, and answer the reflection questions based on the instructor demo. Technical honesty matters.

Suggested 20-Point Rubric

Category	Points
Terminal and navigation evidence	4
File creation, viewing, and search evidence	4
hello.sh script and output	4
system_snapshot.sh script and output	4
Writing/reflection answers	4

Source Alignment

- Project 2-2: BASH shell basics, terminal prompts, case sensitivity, `who/date` commands.
- Project 3-1: `pwd`, `cd`, absolute paths, relative paths, and Linux directory navigation.
- Chapter 3 Linux Directory Structure: file viewing, `cat`, `head`, `tail`, `grep`, `file`, and directory tree concepts.
- Create a BASH Script, Save, Execute: shebang, `chmod +x`, and running a script.